

Mathematics — Reference Handbook

Mathematics (MATH) · Unit 1 · Matrices & Linear Algebra · 2024

Overview

These notes cover **Mathematics**. Engineering and discrete mathematics — linear algebra, calculus, probability, logic and graph theory for computer science. Each section states the core idea, a worked intuition and the points examiners look for.

1. Rank of a Matrix

The rank is the number of linearly independent rows, found by reducing to echelon form.

Why it matters: Rank of a Matrix appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of rank of a matrix in one line.
- A small example showing rank of a matrix in practice.
- Common mistakes students make and how to avoid them.

2. Systems of Linear Equations

Consistency of $AX = B$ is decided by comparing the ranks of A and the augmented matrix.

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3. Eigenvalues and Eigenvectors

Eigenvalues solve $\det(A - \lambda I) = 0$; eigenvectors span the corresponding null spaces.

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- Key definition of eigenvalues and eigenvectors in one line.
- A small example showing eigenvalues and eigenvectors in practice.
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4. Mean Value Theorems

Rolle's, Lagrange's and Cauchy's mean value theorems relate derivatives to function values.

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- Key definition of mean value theorems in one line.
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5. Taylor and Maclaurin Series

Taylor series expand a function as a power series around a point.

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- Key definition of taylor and maclaurin series in one line.
- A small example showing taylor and maclaurin series in practice.
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6. Partial Derivatives

Partial derivatives measure change with respect to one variable while others are fixed.

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- Key definition of partial derivatives in one line.
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7. Conditional Probability

$P(A|B) = P(A \text{ and } B) / P(B)$ gives the probability of A given B occurred.

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- Key definition of conditional probability in one line.
- A small example showing conditional probability in practice.
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8. Bayes' Theorem

Bayes' theorem updates prior probabilities using observed evidence.

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- Key definition of bayes' theorem in one line.
- A small example showing bayes' theorem in practice.
- Common mistakes students make and how to avoid them.

9. Random Variables

Discrete and continuous random variables are described by PMFs, PDFs and CDFs.

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- Key definition of random variables in one line.
- A small example showing random variables in practice.
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10. Propositional Logic

Propositions combine with connectives; truth tables test tautologies and contradictions.

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- Key definition of propositional logic in one line.
- A small example showing propositional logic in practice.
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11. Predicate Logic

Quantifiers for all and there exists extend propositional logic to statements about objects.

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- Key definition of predicate logic in one line.
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12. Rules of Inference

Modus ponens, modus tollens and resolution derive valid conclusions.

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- Key definition of rules of inference in one line.
- A small example showing rules of inference in practice.
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13. Permutations and Combinations

nPr counts ordered arrangements and nCr counts unordered selections.

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- Key definition of permutations and combinations in one line.
- A small example showing permutations and combinations in practice.
- Common mistakes students make and how to avoid them.

14. Pigeonhole Principle

If $n + 1$ objects go into n boxes, some box holds at least two objects.

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- Key definition of pigeonhole principle in one line.
- A small example showing pigeonhole principle in practice.
- Common mistakes students make and how to avoid them.

15. Recurrence Relations

Linear recurrences are solved with characteristic equations or generating functions.

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- A small example showing recurrence relations in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Rank of a Matrix · Systems of Linear Equations · Eigenvalues and Eigenvectors · Mean Value Theorems · Taylor and Maclaurin Series · Partial Derivatives · Conditional Probability · Bayes' Theorem · Random Variables · Propositional Logic · Predicate Logic · Rules of Inference · Permutations and Combinations · Pigeonhole Principle · Recurrence Relations